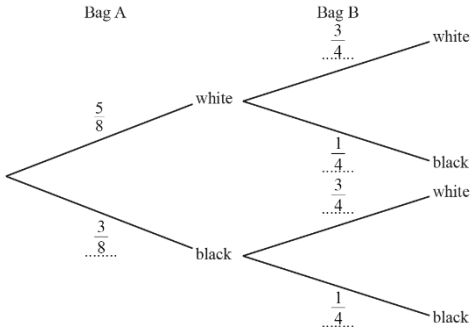
Question	Answer	Marks	Partial Marks
20(a)	0	**1**	
20(b)(i)		**2**	**B1** for $\frac{3}{8}$ or for $\frac{3}{4}$ and $\frac{1}{4}$ correctly placed on the same pair of branches
20(b)(ii)	$\frac{18}{32}$ oe	**3**	**M2** for $\frac{5}{8}\times their\,\frac{3}{4}+their\,\frac{3}{8}\times their\,\frac{1}{4}$ oe or **M1** for one correct branch e.g. $\frac{5}{8}\times their\,\frac{3}{4}$ or $their\,\frac{3}{8}\times their\,\frac{1}{4}$ oe
20(c)	$\frac{11}{40}$ oe	**3**	**M2** for $\frac{5}{8}\times\frac{1}{5}+\frac{3}{8}\times\frac{2}{5}$ oe or **M1** for one product of the form $\frac{5}{8}\times\frac{k}{5}$ or $\frac{3}{8}\times\frac{k}{5}$ oe (k = 1, 2, 3 or 4)
21(a)	[a =] –9 [b =] 7	**2**	**B1** for each or for three correct terms from $6+9\sqrt{5}-2\sqrt{5}-3\sqrt{5}\sqrt{5}$ oe
21(b)	$3\sqrt{2}$ final answer	**2**	**M1** for $\times\frac{\sqrt{2}}{\sqrt{2}}$ oe
22	–1 and 4	**5**	**B2** for $x^2 - 3x - 4$ [= 0] or **M1** for $2(x + 2) = x(x - 1)$ or better **M2** for a correct method to solve *their* three-term quadratic in the numerator or **M1** for $(x + a)(x + b)$ where $ab = -4$ or $a + b = -3$ or $x(x-4)+[1](x-4)$ or $x(x+1)-4(x+1)$ or **M1** for $\sqrt{(-3)^2-4\times[1]\times-4}$ or better or if in the form $\frac{p+\sqrt{q}}{r}$ or $\frac{p-\sqrt{q}}{r}$ then **M1** for $p = -(-3)$ and $r = 2(1)$ or better